

Part A

Question 1:

- a)
i.
Section 1 and 2 are core.

ii.
Section 3 is non-core.

b)

i	1	2	3	4
✓	✓	✓	✓	✓
5	6	7	8	9
✓	✓	✓	✓	✓
10	11	12	13	14
✓	✓	✓	✓	✓
15	16			
✓	✓			

Practice quiz 2

Summary of attempt

Question	Status	Marks
1	Correct	2.00
2	Correct	1.50
3	Correct	0.89
4	Correct	1.00
5	Correct	1.00
6	Correct	0.75
7	Correct	1.00
8	Correct	1.93
9	Correct	2.00
10	Correct	1.00
11	Correct	2.00
12	Correct	2.00
13	Correct	1.00
14	Correct	1.00
15	Correct	0.50
16	Correct	0.50

2/2

Question 2:

$$G = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$$

and $+_G$ be

$$(a + a') + (b + b')\sqrt{-5}$$

and a subgroup of G

$$H = \{a + b\sqrt{-5} : a, b \in 4\mathbb{Z}\}$$

a)

i.

The identity element, e , of G is, $0 + 0\sqrt{-5}$. ✓

ii.

An element of H that is not the identity is, $4 + 8\sqrt{-5}$. ✓

iii.

An element of G that is not in H is, $3 + 5\sqrt{-5}$. ✓

2/2

iv.

The sum, under $+_G$, of these two elements are,

$$\begin{aligned} (4 + 8\sqrt{-5}) + (3 + 5\sqrt{-5}) &= (4 + 3) + (8 + 5)\sqrt{-5} \\ &= 7 + 13\sqrt{-5} \end{aligned} \quad \checkmark$$

b)

G1 Closure:

Let,

$$x = a + b\sqrt{-5} \quad \text{and} \quad y = a' + b'\sqrt{-5}$$

then

$$x + y = (a + a') + (b + b')\sqrt{-5}$$

and as $\mathbb{Z}$ is closed under addition $a + a' \in \mathbb{Z}$ and $b + b' \in \mathbb{Z}$, Hence,
 $x + y \in G$. ✓

G2 Identity element:

Let,

$$e = 0 + 0\sqrt{-5}$$

and

$$x = a + b\sqrt{-5}$$

then

$$\begin{aligned} x + e &= (a + 0) + (b + 0)\sqrt{-5} \\ &= a + b\sqrt{-5} \\ &= x \\ &= 0 + x \end{aligned}$$

G3 Inverse element:

Let,

$$x = a + b\sqrt{-5}$$

then

$$\begin{aligned} x + (-x) &= (a + -a) + (b + -b)\sqrt{-5} \\ &= 0 + 0\sqrt{-5} \\ &= e \\ &= -76 + 76 \end{aligned}$$

Note that elements of G are not themselves integers

3.5/4

G4 Associativity:

The associativity condition holds for addition in $\mathbb{Z}$,

Hence $(G, +_G)$ is a group.

All of the properties are ultimately inherited from $\mathbb{Z}$, but you still have to explain how.

You could argue that associativity holds because we know that addition of complex numbers is associative, and the elements of G are all complex numbers. However, if you want to argue that this follows from associativity of + on the integers there is some extra work to do.

See my email dated 5th December.

c)

Using Proposition 2.5; Subgroup criteria, HB Chapter 5 p26.

You need to mention that by part (a)(ii), H is non-empty. This is part of the criteria.

$$x, y \in H \implies x - y \in H$$

Let,

$$x = 4m + 4n\sqrt{-5} \quad \text{and} \quad y = 4p + 4q\sqrt{-5}$$

then

$$\begin{aligned} x - y &= (4m - 4p) + (4n - 4q)\sqrt{-5} \\ &= 4(m - p) + 4(n - q)\sqrt{-5} \end{aligned}$$

2/2

As $m - p \in \mathbb{Z}$ and $n - q \in \mathbb{Z}$, then $x - y \in H$. Hence, H is a subgroup of G .

d)

The number of distinct cosets of H in G is 16.

- $a \pmod{4}$ has 4 possible values; 0, 1, 2, 3 in $4\mathbb{Z}$.
- $b \pmod{4}$ has 4 possible values; 0, 1, 2, 3 in $4\mathbb{Z}$.

1/1

Thus, by the counting principle, the total number of cosets is $4 \times 4 = 16$.

e)

The order of the element $1 + \sqrt{-5} + H \in \frac{G}{H}$ is 4.

1/1

$$(1 + \sqrt{-5}) + H, \quad 2(1 + \sqrt{-5}) + H, \quad 3(1 + \sqrt{-5}) + H, \quad 4(1 + \sqrt{-5}) + H = H$$

9.5/10

Question 3:

$$M = \begin{pmatrix} a, 0 \\ b, c \end{pmatrix} : a, b, c \in \mathbb{Z}_5$$

a)

The additive inverse is

You are asked about the specific matrix in the question, not a generic element of the group.

$$\begin{pmatrix} -a, -0 \\ -b, -c \end{pmatrix} \quad \times \quad \begin{pmatrix} 4 & 0 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ -2 & -2 \end{pmatrix}$$

0/1

b)

Let

$$\varphi : M \rightarrow \mathbb{Z}_5$$

be the map that takes a matrix in M to the lower left entry in M .*Proof.* To show that φ is a homomorphism, let

no commas in matrix notation $X = \begin{pmatrix} a, 0 \\ b, c \end{pmatrix}$ and $Y = \begin{pmatrix} d, 0 \\ e, f \end{pmatrix}$

then

$$\begin{aligned} \varphi(X + Y) &= \varphi \begin{pmatrix} a + d, 0 + 0 \\ b + e, c + f \end{pmatrix} \\ &= b + e \end{aligned}$$

and

$$\varphi(X) + \varphi(Y) = b + e$$

Thus,

$$\varphi(X + Y) = \varphi(X) + \varphi(Y)$$

Hence, φ is a homomorphism. 

□

3/3

c)

The kernel of φ is,

The kernel is a set, so you need set brackets

$$\text{Ker}(\varphi) = \left\{ \begin{pmatrix} a, 0 \\ 0, c \end{pmatrix} : a, c \in \mathbb{Z}_5 \right\}$$

0/1

because the lower-left entry must be 0.

This doesn't answer the question. You are asked to write down a non-zero element of the kernel. Any specific element will do, e.g. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

d)

A group that is isomorphic to the quotient group $M / \text{Ker}(\varphi)$ is $\mathbb{Z}_5$

1/1

4/6

Question 4:

$$G = \mathbb{Z}_{101} \times \mathbb{Z}_{50}$$

a)

As 101 and 50 are coprime (Corollary 2.12, HB Chapter 6, pg.32),

$$\mathbb{Z}_{101} \times \mathbb{Z}_{50} \cong \mathbb{Z}_{5050}$$

The order of G is $101 \times 50 = 5050$.

i.

The order of the element $(25, 1)$ is;

$$\frac{101}{\text{hcf}(101, 25)} = 101$$

$$\frac{50}{\text{hcf}(50, 1)} = 50$$

Thus, the order of $(25, 1)$ is

$$\text{lcm}(101, 50) = 5050$$

v good
2/2

Hence $(25, 1)$ is a generator of G .

ii.

The order of the element $(1, 25)$ is;

$$\frac{101}{\text{hcf}(101, 1)} = 101$$

$$\frac{50}{\text{hcf}(50, 25)} = 2$$

Thus, the order of $(1, 25)$ is

$$\text{lcm}(101, 2) = 202$$

2/2

Hence $(1, 25)$ is not a generator of G .

b)

i.

First group

$$\mathbb{Z}_{600} \times \mathbb{Z}_{49}$$

The order of this group is $600 \times 49 = 29400$.

Using Corollary 2.12 and Theorem 2.13, HB Chapter 6, p32,

$$\mathbb{Z}_{600} \cong \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_{25}$$

$$\mathbb{Z}_{49} \cong \mathbb{Z}_{49}$$

Hence,

$$\mathbb{Z}_{600} \times \mathbb{Z}_{49} \cong \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_{25} \times \mathbb{Z}_{49}$$

Second group

$$\mathbb{Z}_2 \times \mathbb{Z}_{12} \times \mathbb{Z}_{175} \times \mathbb{Z}_7$$

The order of this group is $2 \times 12 \times 175 \times 7 = 29400$.

Using Corollary 2.12 and Theorem 2.13, HB Chapter 6, p32,

$$\mathbb{Z}_2 \cong \mathbb{Z}_2$$

$$\mathbb{Z}_{12} \cong \mathbb{Z}_4 \times \mathbb{Z}_3$$

$$\mathbb{Z}_{175} \cong \mathbb{Z}_{25} \times \mathbb{Z}_7$$

$$\mathbb{Z}_7 \cong \mathbb{Z}_7$$

Therefore a decomposition as a direct product of cyclic groups of prime power order is,

$$\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_{25} \times \mathbb{Z}_7 \times \mathbb{Z}_7$$

Thus (By Theorem 2.13, HB Chapter 6, p32),

$$\mathbb{Z}_2 \times \mathbb{Z}_{12} \times \mathbb{Z}_{175} \times \mathbb{Z}_7 \cong \mathbb{Z}_{14} \times \mathbb{Z}_{2100}$$

Third group

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{1225}$$

The order of this group is $2 \times 2 \times 6 \times 1225 = 29400$.

Using Corollary 2.12 and Theorem 2.13, HB Chapter 6, p32,

$$\begin{aligned}\mathbb{Z}_2 &\cong \mathbb{Z}_2 \\ \mathbb{Z}_6 &\cong \mathbb{Z}_2 \times \mathbb{Z}_3 \\ \mathbb{Z}_{1225} &\cong \mathbb{Z}_{25} \times \mathbb{Z}_{49}\end{aligned}$$

3/3

Thus (By Theorem 2.13, HB Chapter 6, p32),

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{1225} \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_{25} \times \mathbb{Z}_{49}$$

ii.

By Theorem 2.11, HB Chapter 6, p32, a direct product is cyclic if and only if the component orders are coprime.

- First group; $\mathbb{Z}_{600} \times \mathbb{Z}_{49}$; $\text{hcf}(600, 49) = 1$ so it is cyclic.
- Second group; $\mathbb{Z}_2 \times \mathbb{Z}_{12} \times \mathbb{Z}_{175} \times \mathbb{Z}_7$; $\text{hcf}(2, 12) = 2$ so it is not cyclic.
- Third group; $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{1225}$; $\text{hcf}(2, 6) = 2$ so it is not cyclic.

3/3

Therefore only the group $\mathbb{Z}_{600} \times \mathbb{Z}_{49}$ is cyclic.

c)

Since

$$1144 = 2^3 \times 11 \times 13$$

and, Using Corollary 2.12, HB Chapter 6, p32,

$$\mathbb{Z}_{11} \times \mathbb{Z}_{13} \cong \mathbb{Z}_{143}$$

Two possible (non-isomorphic) non-cyclic abelian groups of order 1144 are,

$$\mathbb{Z}_{1144} \not\cong \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_{143}$$

$$\mathbb{Z}_{1144} \not\cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{143}$$

2/2

They are non-isomorphic because their 2-power components differ.

12/12

Question 5:

$$D_{50} = \langle r, s \mid r^{50} = s^2 = e, sr = r^{49}s \rangle$$

a)

$$\text{Let } H = \{r^{2i}s^j : i = 0, \dots, 24, j = 0, 1\}$$

Using Proposition 2.4, HB p25.

*Proof.***Non-empty:**

$$\text{Let } i = 0, j = 0$$

$$r^0 s^0 = e \in H$$

Thus, H is non-empty.

Closure $x, y \in H \implies x^{-1}y \in H :$

Let

$$x = r^{2a}s^b \text{ and } y = r^{2c}s^d, \quad a, c \in \{0, \dots, 24\}, \quad b, d \in \{0, 1\}$$

then

$$x^{-1} = s^b r^{-2a}$$

Since $sr = r^{-1}s$, we can move s past powers of r .

$$\begin{aligned} x^{-1}y &= r^{-2a}s^b \cdot r^{2c}s^d \\ &= r^{-2a} \cdot s^{b+d} \cdot r^{2c} \end{aligned}$$

Always define all of your notation. What is j? Maybe j=b+d?Let $j = 0$,

$$\begin{aligned} x^{-1}y &= r^{-2a} \cdot r^{2c} \\ &= r^{2(c-a)} \in H \end{aligned}$$

Let $j = 1$,

$$x^{-1}y = r^{-2a}s \cdot r^{2c}s$$

Again since $sr = r^{-1}s$, we can move s past powers of r .

$$= r^{-2a} \cdot r^{2c} \cdot s^2$$

As $s^2 = e$, ✓

$$= r^{2(c-a)} \in H$$

good

Hence, H is a subgroup of D_{50} . ✓

2/3

□

b)

The order of the group D_{50} is 100. The order of the subgroup H is 50.

Hence the index of H in D_{50} is

$$\frac{|D_{50}|}{|H|} = \frac{100}{50} = 2$$

1/1

As the index of H is 2, its left and right cosets are the same, hence H is a normal subgroup of D_{50} , by Proposition 3.7, HB Chapter 6, p27.

Not the right result. You need to quote Exercise 3.3 of Ch 5.

c)

Capital K

Let k be the subgroup generated by r^{25} .

The order of k is 2, as

$$\begin{aligned} (r^{25})^1 &= r^{25} \\ (r^{25})^2 &= r^{50} = e \end{aligned}$$

1/1

Which is isomorphic to $\mathbb{Z}_2$ which also has order 2.

You should say that $\mathbb{Z}_2$ is the only group of order 2 up to isomorphism, and as such k must be isomorphic to $\mathbb{Z}_2$.

d)

Given $D_{50} \langle sr = r^{49}s = r^{-1}s \rangle$;

$$Z(H) = \{a \in H : ah = ha \forall h \in H\}$$

$$sr^n = r^{-n}s$$

$$sr^{25} = r^{-25}s$$

$$= r^{50-25}s$$

$$= r^{25}s$$

Since $sr^{25} = r^{-25}s = r^{25}s$ (because $r^{50} = e$), r^{25} commutes with both r and s and so lies in the centre. By Exercise 2.5 of chapter 7, subgroups of $Z(G)$ are normal subgroups of G , thus K is normal. ✓

good
3/3

Hence, by Proposition 3.7, HB chapter 6 p27, k is a normal subgroup of D_{50} .

e)

All elements in the subgroup H are of the form $r^{2i}s^j$, with the exponent of the r term being even and the element e .

1/1

As r^{25} has an odd exponent, its only other element is e .

Thus $H \cap k = \{e\}$.

f)

By Proposition 1.9, HB Chapter 6 p31,

D_{50} has order 100

H has order 50

k has order 2

Say why part (b) of the proposition is satisfied.

3/3

As $100 = 50 \times 2$, and $H \cap k = \{e\}$, then $D_{50} \cong H \times k$.

11/12

Question 6:

Given G is a group of order $48668 = 4 \times 23^3$.

a)

By Definition 1.7, HB Chapter 8 p38, the Sylow 2-subgroups of G are order 4, and the Sylow 23-subgroups of G are order $23^3 = 12167$.

1/1

b)

Using The Sylows Theorems, Theorem 1.12, HB Chapter 8 p39,

For the Sylow 2-subgroups of G ;

$$\begin{aligned} n_2 &\equiv 1 \pmod{2} \\ n_2 \mid 12167 &\equiv 1 \pmod{2} \end{aligned}$$

Define your notation. The usual notation is m_2 , etc. Using n_2 is fine, but you have to introduce it.

given the factors of 12167 are 1, 23, 529, 12167, thus the possible values of n_2 are;

$$n_2 = 1, 23, 529, 12167$$

Tell me which Sylow Theorems you are using for which parts. In the exam you need precise references to gain full marks.

as all these possible values are odd, they are all congruent to 1 (mod 2).

For the Sylow 23-subgroups of G ;

$$n_{23} \mid 4 \equiv 1 \pmod{23}$$

This notation doesn't make sense. You have to write this as two conditions.

given the factors of 4 are 1, 2, 4, thus the possible value of n_{23} are;

$$1 \equiv 1 \pmod{23}$$

$$2 \equiv 2 \pmod{23}$$

$$4 \equiv 4 \pmod{23}$$

3.5/4

as 1 is the only value congruent to 1 (mod 23), thus by Lemma 1.9, HB Chapter 8 p39, there is a unique Sylow 23-subgroup of G , which is normal.

4.5/5

Question 7:**a)**

Let q be a prime, with $q \neq 29$

A counter example to the proposition that the order of the group $29q$ is cyclic is;

$$q = 2, \text{ with } |29q| = 58$$

1/1

and thus,

D_{29} is non abelian, thus not cyclic.

b)

Line 7 of the 'proof' is false. Using Definition 3.6, Propositions 3.7 and 3.10, HB Chapter 5 p 27, H and K are cyclic subgroups of G , but being cyclic does not imply being normal. A cyclic subgroup of a non-abelian group need not be normal.

2/2

3/3

Part B

Question 8: Question 9 from TMA02 material.

Let $C[0,1]$ be the group of all continuous functions from $[0,1]$ to $\mathbb{R}$. Let $+$ be the operation defined By

$f + g$ is the function from $[0,1]$ to $\mathbb{R}$ given by $(f + g)(x) = f(x) + g(x)$

G1 Closure: If $f, g \in C[0,1]$ then $x \in C[0,1], f(x), g(x) \in \mathbb{R}$

Since the sum of two continuous functions is continuous, Combination rules for continuous functions from $\mathbb{R}$ to $\mathbb{R}$, HB Chapter 13 p65,

$$f + g \in C[0,1]$$

Hence, $C[0,1]$ is closed under $+$.

G2 Identity: Let e be the function from $[0,1]$ to $\mathbb{R}$ defined by $e(x) = 0$ for all $x \in [0,1]$.

Then for any $f \in C[0,1]$ and $x \in [0,1]$,

$$\begin{aligned} (f + e)(x) &= f(x) + e(x) \\ &= f(x) + 0 \\ &= f(x) \end{aligned} \quad \text{Remember to check both ways (or comment that the operation is commutative).}$$

Thus, $f + e = f$ for all $f \in C[0,1]$, so e is the identity element of $C[0,1]$.

G3 Inverse: Define $(-f)(x) = -f(x)$, for all $x \in [0,1]$.

Since f is continuous $-f$ is also continuous, by Combination rules for continuous functions from $\mathbb{R}$ to $\mathbb{R}$, HB Chapter 13 p65.

Then

$$\begin{aligned} (f + (-f))(x) &= f(x) - f(x) \\ &= 0(x) = e \end{aligned} \quad \text{Remember to check both ways (or comment that the operation is commutative).}$$

Thus every element has an inverse in $C[0,1]$.

G4 Associativity: Associativity holds because we are told in the question that addition in $\mathbb{R}$ is associative. **You still need to show the connection - see the specimen solution.**

Hence, $(C[0, 1], +)$ is a group.

6/8

Question 9: Question 11 from TMA02 material.

a)

$$\mathbb{Z}_3 \times \mathbb{Z}_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_6 \quad \times$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \cong \mathbb{Z}_4 \quad \times$$

These are not isomorphic because RHS is not cyclic, but the RHS is (since 3 and 4 are coprime, so RHS is isomorphic to $\mathbb{Z}_{12}$).

These are not isomorphic because LHS is not cyclic (2 is not coprime with 2) but the RHS is.

0/2

A non-abelian group cannot be isomorphic to an abelian one: isomorphisms preserve all group theoretic properties.

b)

Group	Order of group	Isomorphic groups	Order of elements	Cyclic? Abelian
$\mathbb{Z}_{15}$	15	$\mathbb{Z}_3 \times \mathbb{Z}_5$	1, 3, 5, 15	Cyclic
$\mathbb{Z}_2 \times \mathbb{Z}_{10}$	20	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5$	1, 2, 5, 10	Abelian, non-cyclic
Dic_6	24 ✓	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_6 \quad \times$	1, 2, 3, 4, 6, 12, 24 ✓	non-abelian ✓
$\mathbb{Z}_3 \times \mathbb{Z}_4$	12 ✓	$\mathbb{Z}_{12} \quad \checkmark$	1, 2, 3, 4, 6, 12 ✓	Cyclic, Abelian ✓
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3$	12 ✓	$\mathbb{Z}_{12} \quad \times$	1, 2, 3, 4, 6 ✓	Non-Cyclic, abelian.

8/12

No part (c): 0/2

8/16